

Assignment: Day 4

Hands-on Subnetting

Q1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Ans: To get 4 equal subnets I borrowed 2 bits from the host part, so the new prefix is **/26** and the subnet mask for every subnet is **255.255.255.192**. Each subnet has 64 addresses (62 usable).

Subnet	Subnet mask	Network address	Usable host range	Broadcast address
Subnet 1	255.255.255.192	192.168.10.0/26	192.168.10.1 to 192.168.10.62	192.168.10.63
Subnet 2	255.255.255.192	192.168.10.64/26	192.168.10.65 to 192.168.10.126	192.168.10.127
Subnet 3	255.255.255.192	192.168.10.128/26	192.168.10.129 to 192.168.10.190	192.168.10.191
Subnet 4	255.255.255.192	192.168.10.192/26	192.168.10.193 to 192.168.10.254	192.168.10.255

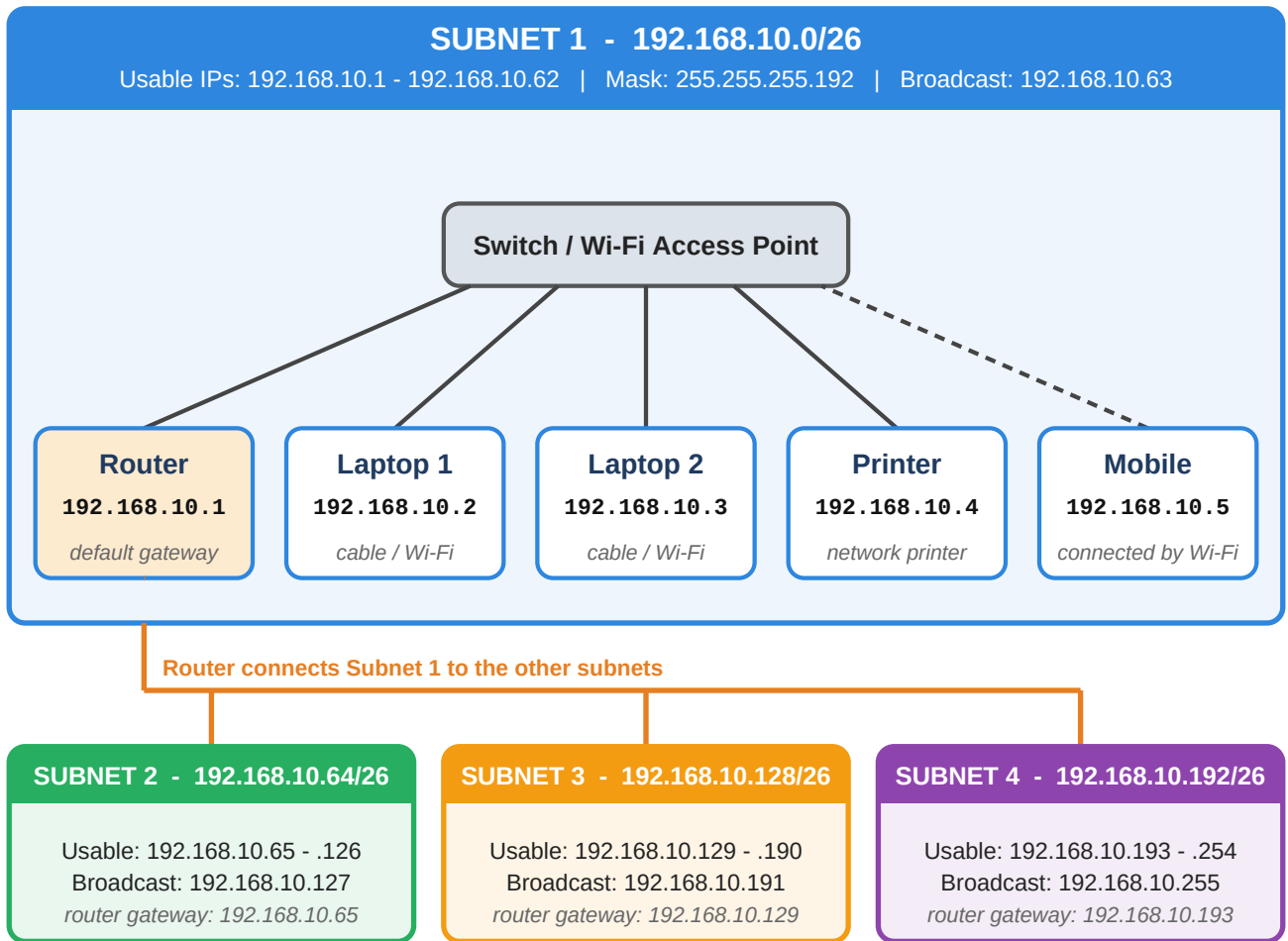
Q2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Ans: I chose **Subnet 1 (192.168.10.0/26)**. Its network address (192.168.10.0) and broadcast address (192.168.10.63) are not used for any device. All devices use the mask 255.255.255.192 and the router as the default gateway.

Device	IP address	Subnet mask	Default gateway
Router (default gateway)	192.168.10.1	255.255.255.192	-
Laptop 1	192.168.10.2	255.255.255.192	192.168.10.1
Laptop 2	192.168.10.3	255.255.255.192	192.168.10.1
Printer	192.168.10.4	255.255.255.192	192.168.10.1
Mobile	192.168.10.5	255.255.255.192	192.168.10.1

Q3. Draw a simple diagram showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

Ans:



Q4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Ans: Think of 192.168.10.0/24 as a chocolate bar with 256 pieces. Subnetting simply means cutting it into equal blocks. Here is how I did it, step by step.

Step 1: Find out how many addresses we have.

The /24 means the first 24 bits are the network part. An IPv4 address has 32 bits, so $32 - 24 = 8$ bits are left for hosts. That gives $2^8 = 256$ addresses in total.

Step 2: Decide how many subnets we need.

We need 4 equal subnets. To make subnets we borrow bits from the host part. Each borrowed bit doubles the number of subnets: 1 bit = 2, 2 bits = 4. So I need to borrow **2 bits** ($2^2 = 4$).

Step 3: Find the new prefix.

Borrowing 2 bits makes the network part longer: $24 + 2 = \mathbf{/26}$.

Step 4: Write the subnet mask.

A /26 mask has 26 ones followed by 6 zeros: 11111111.11111111.11111111.11000000. In the last octet, $11000000 = 128 + 64 = 192$. So the subnet mask is **255.255.255.192**.

Step 5: Count the hosts in each subnet.

Host bits left = $32 - 26 = 6$. So each subnet has $2^6 = 64$ addresses. Two of them cannot be given to devices (the first is the network address, the last is the broadcast address), so usable hosts = $64 - 2 = \mathbf{62}$.

Step 6: Find where each subnet starts.

The block size is $256 - 192 = \mathbf{64}$. So the subnets start at 0, 64, 128 and 192. The network address is the first address of a block and the broadcast address is the last one (the next network address minus 1). For example, Subnet 1 runs from 192.168.10.0 to 192.168.10.63, so the broadcast address is 192.168.10.63 and the usable range is .1 to .62. The same method gives the other three subnets in the table of Q1.

Step 7: Assign IP addresses to devices.

I picked Subnet 1. I skipped 192.168.10.0 (network) and 192.168.10.63 (broadcast). I gave the router the first usable address, 192.168.10.1, because the router is the gateway and it is easy to remember. Then I gave the next addresses to the other devices: Laptop 1 = .2, Laptop 2 = .3, Printer = .4, Mobile = .5.

Step 8: Check the plan.

Every device address is between .1 and .62, no two devices share an address, and all of them use the mask 255.255.255.192 with the gateway 192.168.10.1. Because they are all in the same subnet, they can talk to each other directly, and the router is used only to reach other subnets or the internet.

Quick binary view of the mask (last octet):

Bit value	128	64	32	16	8	4	2	1	Total
Mask bits	1	1	0	0	0	0	0	0	192
Meaning	net	net	host	host	host	host	host	host	